Mastery-Conditioned Edge–Cloud Model Selection for Wearable Psychomotor Skill Learning

A Case Study in Network Engineering Education

Principal Investigator: Ajay
Affiliation: Faculty of Information Science and Technology (FIST), Multimedia University, Melaka
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0. One-Sentence Contribution

> Offloading decisions in wearable skill-training systems should depend on **how much the learner has mastered**, not only on how uncertain the model is — and compute demand can be shown to decay measurably along the skill-acquisition curve.

Everything in this document exists to defend, implement, or falsify that sentence. If a section does not serve it, cut the section.

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1. Background and Motivation

Practical psychomotor competence is the terminal outcome of technical and vocational ICT education, yet it is assessed almost entirely by human observation: an instructor watches a student terminate a cable, ticks a rubric, and assigns a grade. This is expensive, low-frequency, subjective, and — critically — provides no formative feedback *during* the attempt, which is when feedback most affects motor learning.

Wearable and instrumented-tool sensing offers a route to continuous, objective, in-situ assessment. Foundation models (FMs) offer the open-set generalisation needed to interpret unconstrained learner behaviour. The two do not compose naturally: FMs are too large for wearables, and cloud offloading of every inference is infeasible on latency, cost, privacy, and connectivity grounds in a real teaching laboratory.

Edge–cloud collaborative systems address exactly this tension. The current state of the art routes between a customised small edge model and a cloud FM using **data uncertainty and network conditions**. This proposal argues that in an *educational*

deployment, a third and more informative variable is available and currently unused: the **learner's mastery state**.

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2. Problem Statement

Existing edge–cloud inference systems treat every input as an independent sample drawn from a stationary process. They select a model by asking: *how uncertain is this sample, and what bandwidth is available right now?*

This assumption fails in a skill-learning deployment for three reasons.

(P1) The input distribution is non-stationary by design. The entire purpose of the intervention is to change the learner. A novice's motion signature is high-variance, error-dense, and hesitant; the same learner six weeks later produces low-variance, fluent, largely correct execution. A routing policy tuned on early-session data is systematically miscalibrated on late-session data, and vice versa. Treating the learner as a stationary source discards the strongest predictive signal in the system.

(P2) The value of escalation is not equal across learners or moments. Uncertainty-triggered offloading assumes that resolving uncertainty is always worth the same. Pedagogically it is not. Escalating to a large model for a learner who has already automated a step yields a confirmation of something already known. Escalating for a learner stalled at a mastery boundary yields the diagnostic explanation that unblocks acquisition. Identical uncertainty, radically different educational return.

(P3) A shared edge node serves many learners under contention. Current systems route a single stream on a single device. A teaching laboratory has one edge node, one uplink, and 20–30 concurrent learners. When simultaneous low-confidence samples exceed the offload budget, existing work provides no principled allocation rule. First-come-first-served and highest-entropy are both pedagogically arbitrary.

Consequently: there is no inference-placement policy for wearable skill-training systems that is conditioned on the learner's acquisition state, and no evidence as to whether such conditioning improves either system efficiency or learning outcomes.

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3. Knowledge Gap

3.1 What exists

Line of work	What it establishes	What it does not do
Edge–cloud FM collaboration	Selective FM querying, semantic customisation of edge models from	No notion of a user who improves; single-

Line of work	What it establishes	What it does not do
(EdgeFM / DeepEdgeFM)	unlabelled data, runtime switching on data uncertainty + network variation, architecture zoo with pre-profiled accuracy/latency/resource cost	stream; no multi-user contention; no educational objective
Wearable psychomotor skill assessment (IMU, gaze, sEMG)	Novice–expert discrimination is reliably achievable; gaze is a strong expertise signal; repetition consistency is a usable mastery proxy	Predominantly post-hoc *scoring*, not in-loop formative intervention; predominantly surgery, sport, welding, aviation — ICT/TVET is near-absent
Gaze-based skill assessment on smart glasses (SkillSight)	Efficient on-glasses skill assessment via distillation with explicit per-sensor power accounting	Assessment as the terminal output; no edge–cloud placement policy; no mastery-conditioned resource allocation
Learner modelling (BKT / DKT / knowledge tracing)	Mature latent-mastery estimation over decomposed skill steps	Almost exclusively applied to cognitive/item-response data, not continuous psychomotor sensor streams; never used as a *systems* control signal
Instructional design (task analysis, chaining, prompt fading)	Validated decomposition and prompt-fading procedures with independence as the outcome measure	No computational instantiation tied to inference cost

3.2 The gap, stated precisely

Learner modelling and edge–cloud inference placement have developed as unconnected literatures. Knowledge tracing produces a continuously-updated latent competence estimate. Edge–cloud systems require a signal on which to condition model selection. **No published work uses the former as an input to the latter.** The consequence is that educational edge systems either over-provision (cloud FM for every inference, unaffordable at cohort scale) or under-provision (fixed small edge model, inadequate diagnostic depth for novices).

3.3 Three sub-gaps this project addresses

- **G1 — Signal gap:** whether a mastery latent estimated from wearable psychomotor streams is stable and informative enough to serve as a control signal for model selection.
- **G2 — Policy gap:** whether conditioning model selection on mastery improves the cost–benefit frontier relative to uncertainty-only and network-only baselines.
- **G3 — Allocation gap:** how to allocate a bounded shared offload budget across concurrent learners in a way that is both pedagogically justified and fair.

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4. Research Questions and Objectives

4.1 Research Questions

- **RQ1.** Can a mastery state over decomposed psychomotor task steps be estimated from wearable and instrumented-tool streams with sufficient accuracy and temporal stability to condition inference placement?
- **RQ2.** Does mastery-conditioned model selection reduce cloud-offload rate, end-to-end latency, and energy per session relative to uncertainty-only routing, **without** degrading assessment accuracy or formative feedback quality?
- **RQ3.** Under multi-learner contention on a bounded offload budget, does a mastery-boundary allocation policy improve aggregate learning gain relative to entropy-ranked and FIFO allocation, and at what cost in inter-learner fairness?
- **RQ4.** Does the resulting system produce measurable improvement in skill acquisition (steps-independent, time-to-criterion, error persistence) relative to conventional instructor-only practice?

4.2 Objectives

ID	Objective	Addresses	Primary output
O1	Construct a task-analytic decomposition and instrumented data-collection pipeline for three network-engineering psychomotor tasks	G1	Annotated multimodal dataset + task graphs
O2	Develop and validate a dual-channel assessment model separating *process quality* from *outcome correctness*	G1, RQ1	Assessment models + validation against expert rubric
O3	Formulate and implement a mastery-conditioned model-selection policy over a pre-profiled architecture zoo	G2, RQ2	Routing policy + calibrated uncertainty layer

ID	Objective	Addresses	Primary output
O4	Extend the policy to bounded-budget multi-learner allocation with an explicit fairness constraint	G3, RQ3	Scheduling algorithm + fairness analysis
O5	Evaluate the integrated system in a live teaching laboratory over a full trimester	RQ4	Field-trial results, learning-gain analysis

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5. Scope and Delimitations

5.1 In scope

- **Skill domain:** psychomotor, procedural, decomposable network-engineering tasks.
- **Target tasks (three, fixed):**
  1. **T1 — RJ45 termination** to T568B, including untwist discipline, pin ordering, jacket seating, crimp, and continuity verification.
  2. **T2 — Fibre preparation** — strip, clean, cleave, inspect, splice/connectorise.
  3. **T3 — Patch-panel punch-down** and structured cable dressing.
- **Population:** undergraduate and matriculation-level learners in networking practicals; adults and near-adults, all consenting.
- **Setting:** supervised teaching laboratory only.
- **Inference placement:** on-body → personal edge → laboratory edge node → cloud FM.

5.2 Explicitly out of scope

These were considered and deliberately excluded. Recording them prevents scope drift and pre-empts reviewer questions.

Excluded	Reason
Clinical / patient-recovery guidance	Triggers Software-as-a-Medical-Device regulation; no clinical standing
Hygiene or ADL training in private spaces	Egocentric capture in bathrooms is not ethically approvable
Minors as participants	Consent/assent tier and device fit make it infeasible at this stage
Cognitive or conceptual learning analytics	Wearables cannot sense conceptual understanding; claim would be unfalsifiable

Excluded	Reason
At-risk / dropout prediction	Runs on a multi-day timescale; no latency justification for edge placement
Facial-affect-based engagement detection	Construct validity of facial-configuration→emotion inference is contested
Unsupervised or home deployment	Safety and data governance

5.3 Assumptions

- Instructors will author and approve the task decomposition; the system does not infer curriculum autonomously.
- Ground-truth outcome correctness is obtainable independently (cable tester, OTDR, visual inspection) for every trial.
- Laboratory Wi-Fi is variable but present; full offline operation is a degraded mode, not the primary mode.

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6. Theoretical Framework

Four theoretical commitments; each maps to a computational component.

6.1 Fitts & Posner three-stage motor learning (cognitive → associative → autonomous).
Provides the referent for the mastery latent. Rather than an unnamed scalar, mastery is anchored to observable stage signatures: high movement variability and heavy visual guidance (cognitive); reduced error rate with persisting inefficiency (associative); low variance, economical gaze, resistance to dual-task interference (autonomous). *Computational role: defines the state space of the learner model and the expected direction of the acquisition trajectory.*

6.2 Task analysis and chaining.
Complex procedures decompose into discrete, individually-assessable steps taught to an independence criterion. *Computational role: defines the skill graph over which knowledge tracing operates, and defines the unit of feedback.*

6.3 Prompt hierarchy and systematic fading.
Support is ordered by intrusiveness and withdrawn as competence rises; failure to fade produces prompt dependency, where the learner waits for the cue despite possessing the skill. *Computational role: defines the ordinal output space of the intervention, and supplies the pedagogical justification for the central systems claim — as prompts fade, required model capacity falls.*

6.4 Knowledge tracing (BKT / DKT).
Latent mastery per step, updated from noisy observations with explicit slip and guess parameters. *Computational role: the estimator itself; the slip/guess structure is what prevents a single sensor misread from collapsing the mastery estimate.*

Synthesis — the coupling claim. Prompt fading (6.3) and stage progression (6.1) both predict that the *information required per intervention* decreases monotonically with mastery. A novice requires error localisation and explanation; an autonomous performer requires only verification. Model capacity requirement therefore inherits the shape of the acquisition curve. This is the theoretical basis for the systems contribution, and it is what makes the efficiency claim a *prediction* rather than an empirical accident.

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7. System Architecture

Four tiers. The tier boundary is set by thermal and power envelope, not by convenience.

```
TIER 0  ON-BODY SENSING
        Wrist IMU (dominant + non-dominant) · eye-tracking glasses
        Role: capture, timestamp, gate. NO inference beyond activity
gating.
              |
TIER 1  PERSONAL EDGE  (per learner)
        Android handset / SBC at the bench
        Role: windowing, feature extraction, tiny models (T-class zoo),
              on-device buffering. Raw video terminates here.
              |
TIER 2  LABORATORY EDGE NODE  (shared, 1 per lab)
        Jetson AGX Orin class
        Role: mid-capacity zoo (S/M-class), mastery estimator for the
cohort,
              contention scheduler, instructor dashboard, session store.
              |
TIER 3  CLOUD FOUNDATION MODEL
        Role: OFFLINE ONLY — semantic customisation of edge models from
              unlabelled session data, open-set error explanation,
              remediation-language generation, zoo re-profiling.
              Never in the hard real-time loop.
```

Instrumented tools (crimper, cable tester, fibre cleaver, punch-down tool) report to Tier 2 directly over BLE/MQTT. They are the **outcome-correctness** channel and are architecturally separate from the on-body **process-quality** channel — see §9.2.

7.1 Design constraints (non-negotiable)

- **C1.** Raw video frames terminate at Tier 1. Only derived representations (hand keypoints, gaze AOI codes, object-contact events, step tokens) cross to Tier 2 or above.
- **C2.** Cloud is never on the critical path for learner-facing feedback. A total uplink failure degrades feedback granularity; it does not stop the session.
- **C3.** The instructor is in the loop by construction — authors the task graph, sets fading criteria, receives live cohort state, can override any system output.
- **C4.** Feedback latency budget: ≤ 300 ms for step-completion confirmation, ≤ 1.5 s for corrective prompts. Beyond ~2 s the learner has already moved on and the feedback attaches to the wrong action.

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8. Hardware and Software Specification

8.1 Hardware

Tier	Component	Indicative spec	Qty	Purpose
0	Wrist IMU	9-axis, ≥200 Hz, BLE 5.x, ≥6 h active	2/learner	Motion, jerk, smoothness, repetition consistency
0	Eye-tracking glasses	≥50 Hz gaze, scene cam, tethered acceptable	4–6 shared	Fixation economy, AOI dwell, quiet-eye
0	sEMG band *(optional, Phase 2)*	2–4 channel forearm	2–4	Grip force, co-contraction (novice marker)
1	Personal edge	Android 13+, ≥8 GB RAM, NPU-capable; or Raspberry Pi 5 + Hailo-8L	1/bench	On-device T-class inference
2	Laboratory edge node	Jetson AGX Orin 64 GB (or Orin NX 16 GB for a reduced cohort)	1/lab	S/M-class zoo, mastery estimator, scheduler
2	Instrumented tools	Crimper with force/closure sensing; cable tester with	6–8 sets	Outcome-correctness ground truth

Tier	Component	Indicative spec	Qty	Purpose
		logging; cleaver with cycle counter		
2	Network	Managed AP with `tc` traffic shaping for controlled bandwidth conditions	1	Reproducible network-variation experiments
—	Power measurement	Inline USB-C PD meter + Jetson `tegrastats`	2	Energy-per-session claims

Note: the eye trackers are the budget-dominant item. If funding is constrained, Phase 1 can proceed with IMU + instrumented tools only, and gaze can be added at Phase 3. Do **not** cut the instrumented tools — without them there is no outcome-correctness channel and the whole assessment claim collapses.

8.2 Software

Layer	Stack
Sensor ingest	BLE GATT → MQTT (Mosquitto); NTP/PTP time sync across all sources
Stream processing	Python 3.11, `river` for online updates, sliding-window segmentation
On-device inference	ExecuTorch / TFLite (Tier 1); TensorRT + ONNX Runtime (Tier 2)
Vision	MediaPipe hand/pose; CLIP-family contrastive matching for step verification
Learner model	Custom BKT (`pyBKT` -derived) and DKT (PyTorch) over the task graph
Uncertainty	`torchuq` / custom conformal wrapper; temperature scaling as floor
Cloud FM	API-based VLM for offline customisation and remediation text
Orchestration	Docker Compose on Tier 2; MLflow for zoo profiling and experiment tracking
Storage	TimescaleDB (streams), MinIO (derived artefacts), all on-premise

Layer	Stack
Privacy	On-device DP-SGD (Opacus) for federated model updates across labs
Dashboard	React + WebSocket; instructor cohort view

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9. Algorithms

9.1 Architecture zoo

A pre-profiled set of models, each benchmarked offline for accuracy, latency, memory, and energy on its target tier — mirroring the DeepEdgeFM zoo approach but extended with an educational utility profile.

Class	Tier	Example	Role
T (tiny)	1	1D-CNN / TCN over IMU windows	Step segmentation, binary step-completion
S (small)	2	Multimodal TCN + gaze encoder	Step classification + process-quality scoring
M (medium)	2	Quantised VLM (2–3 B), CLIP-style step verification	Outcome verification, error localisation
L (large)	3	Cloud FM	Open-set error explanation, remediation generation, distillation source

9.2 Dual-channel assessment

Deliberately separated; conflating them is the principal methodological trap in this literature.

- **Process quality** `Q_p` — from IMU/gaze/sEMG. Movement smoothness (spectral arc length or dimensionless jerk), inter-trial repetition consistency, fixation count and dwell distribution, co-contraction index. Continuous, [0,1].
- **Outcome correctness** `C_o` — from instrumented tools and Tier-1 vision at commitment moments. Binary or categorical per step.

Both feed the mastery estimator. Their **disagreement is diagnostically valuable**: high `Q_p` with low `C_o` indicates a *stabilised error* — fluent, automated, and wrong. This is

the highest-priority intervention state and the system must flag it explicitly, because conventional fluency-based assessment systematically misses it.

9.3 Mastery estimator

Per-step latent mastery `m_{s,k} ∈ [0,1]` for learner `s`, step `k`, updated online.

- BKT baseline with explicit slip/guess; DKT variant for comparison.
- **Engagement latent `e_s`** carried alongside mastery to prevent conflating *cannot* with *did not attempt*. Prompt re-escalation is gated on `e_s`.
- Temporal smoothing to prevent single-trial sensor noise from driving routing thrash.

9.4 Calibration layer

Raw softmax confidence is not a usable router — neural classifiers are systematically overconfident and confidence quality degrades under the exact distribution shift this deployment creates by design.

- Split-conformal prediction for distribution-free coverage guarantees on step classification.
- Temperature scaling as the minimum baseline for comparison.
- **Recalibration checkpoint at each phase boundary**, since the learner distribution is non-stationary. This is a required experimental control, not an optimisation.

9.5 Mastery-conditioned selection policy — the core contribution

For learner `s` at step `k`, choose model class `a ∈ {T,S,M,L}` maximising expected **educational utility per unit system cost**:

```
a*(s,k) = argmax_a [ U(a, m_{s,k}, e_s, r_k) − λ · Cost(a, net, energy)
]

subject to:
  (i)   PRIVACY GATE   — raw-modality egress prohibited (hard,
lexicographic)
  (ii)  LATENCY GATE   — E[latency(a)] ≤ budget(feedback_type)
  (iii) SAFETY GATE    — risk-weighted uncertainty on critical steps r_k
```

Gates are evaluated **before** the utility optimisation, in order. A privacy constraint expressed as a soft weight can be bought out by a sufficiently large bandwidth term; expressed as a lexicographic gate, it cannot.

Utility `U` is designed to peak where mastery is *intermediate* — at the acquisition boundary. Low mastery on an unattempted step yields little from a large model (the learner needs instruction, not diagnosis); high mastery needs only verification. The predicted consequence, and the falsifiable claim of this project, is that **offload rate follows an inverted-U over the acquisition curve and then decays monotonically.**

9.6 Multi-learner contention scheduler

Given `N` concurrent learners and offload budget `B` (concurrent cloud/M-class slots):

- Rank by **marginal expected learning gain** from escalation, not by entropy and not FIFO.
- Enforce a fairness floor: a minimum guaranteed escalation rate per learner per session, so that consistently-high-mastery or consistently-low-engagement learners are not starved.
- Report the efficiency–fairness trade-off curve explicitly. Reviewers will ask; having the curve pre-computed is a strength.

Baselines for comparison (all four must be implemented):
`B1` cloud-only · `B2` edge-only fixed model · `B3` uncertainty + network routing (DeepEdgeFM-equivalent reimplementation) · `B4` proposed mastery-conditioned routing.

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10. Input / Output Specification

10.1 Inputs

Stream	Source	Rate	Representation
Wrist motion	IMU ×2	200 Hz	9-axis, windowed 2 s / 50 % overlap
Gaze	ET glasses	50–100 Hz	Fixations, AOI codes, dwell, saccade amplitude
Scene	Glasses cam	30 fps	**Terminates Tier 1**; exports contact events + step tokens only
Muscle activation	sEMG (opt.)	1 kHz	RMS envelope, co-contraction index
Tool telemetry	Instrumented tools	Event-driven	Crimp closure, tester pass/fail + pin map, cleave angle
Task context	Instructor console	Session	Task ID, task graph version, fading criteria
System context	Tier 1/2 agents	1 Hz	Bandwidth, RTT, battery, thermal, queue depth

10.2 Outputs

To the learner (real time): step-completion confirmation; corrective prompt at the current fading level; stabilised-error alert; end-of-trial summary.

To the instructor (real time + post-session): cohort mastery heatmap (learner × step); learners at a mastery boundary; stabilised-error flags with evidence clip reference; per-step class-wide difficulty ranking.

To the researcher (logged): every routing decision with its full state vector, chosen class, realised latency and energy, and post-hoc counterfactual — *what each unchosen class would have returned.* This counterfactual log is what makes the policy evaluation possible; instrument it from day one.

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11. Implementation Plan — Phases and Timeline

12 months, Sept 2026 – Aug 2027.

Phase	Months	Activities	Gate to proceed
P0 — Foundation	M1–M2	Ethics/IRB submission; task analysis with instructors for T1–T3; expert rubric development; procure Tier 0/1 hardware; reproduce EdgeFM/DeepEdgeFM baseline (`B3`)	Ethics approval **in hand**; `B3` reproduced within 10 % of published figures
P1 — Data & instrumentation	M3–M4	Build ingest pipeline + time sync; instrument tools; collect expert reference set (n≈8 instructors/technicians, ≥20 trials each); collect novice baseline (n≈30)	≥95 % stream sync integrity; expert–novice separation demonstrable on `Q_p`
P2 — Assessment models	M5–M6	Train/validate dual-channel models; zoo profiling (accuracy/latency/energy per class per tier); calibration layer	Step classification ≥85 % on held-out learners; conformal coverage within ±3 % of nominal
P3 — Learner model	M6–M7	BKT/DKT over task graphs; engagement latent; validate mastery trajectory against instructor rubric	Mastery estimate correlates with expert rubric at r ≥ 0.7

Phase	Months	Activities	Gate to proceed
P4 — Routing policy	M7–M9	Implement policy + gates; contention scheduler; simulated-cohort evaluation under replayed traces and shaped network conditions	`B4` dominates `B3` on offload rate at equal accuracy, in simulation
P5 — Field trial	M9–M11	Live deployment across one full trimester of networking practicals; longitudinal acquisition curves	Trial completes with ≥25 learners retained across ≥6 sessions
P6 — Analysis & writing	M11–M12	Results analysis; ablations; two manuscripts	Submission

Critical-path dependencies: Ethics approval gates P1 (do not build sensing before approval). Expert reference data gates P2 (no ground truth, no model). The trimester calendar gates P5 — **if the P5 window is missed, the field trial slips a full trimester.** Work backwards from the practical-class schedule and treat it as immovable.

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12. Expected Results

Stated as directional predictions so they can fail.

#	Prediction	Falsifies if
E1	Mastery-conditioned routing reduces cloud-offload rate by 35–60 % vs. `B3` at statistically equivalent assessment accuracy	Offload reduction < 15 %, or accuracy drop > 2 pp
E2	Offload rate per learner declines monotonically across sessions after an early peak (inverted-U)	No systematic trend, or trend not tied to mastery
E3	End-to-end feedback latency reduced ≥30 % vs. `B3`; energy per session reduced ≥25 %	Improvements within measurement noise
E4	Mastery estimate correlates with instructor rubric at $r \geq 0.7$	$r < 0.5$ — the control signal is not trustworthy and the policy has no basis
E5	Stabilised-error detection (high `Q_p` / low `C_o`) identifies cases instructors missed in ≥1 in 5 occurrences	No stabilised errors detected, or all already caught by instructors

#	Prediction	Falsifies if
E6	Learners using the system reach the independence criterion in fewer trials than the instructor-only comparison group	No difference in time-to-criterion

E2 is the paper. E1 and E3 are systems results obtainable by many means; E2 is the specific signature of *mastery* conditioning and cannot be produced by uncertainty routing alone. Design the analysis around E2.

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13. Validation and Verification

Three independent layers. Conflating them is a common and fatal review finding.

13.1 System verification (does it work as specified?)

- Latency: per-tier profiling; p50/p95/p99 against the §7.1 budgets under shaped network conditions (`tc`: 1/5/20/100 Mbps, 0–200 ms RTT, 0–5 % loss).
- Energy: inline power metering per tier; joules per session and per inference.
- Correctness of routing: assert gate ordering — privacy gate must never be bypassed; adversarial replay to confirm.
- Reproducibility: containerised, seeded, ≥5 runs per configuration, variance reported.

13.2 Model validation (is the inference sound?)

- **Cross-learner** splits, never random splits. Random splits leak the same learner's trials across train/test and inflate accuracy substantially. This is the single most common invalidating error in wearable skill-assessment papers.
- Leave-one-learner-out and leave-one-session-out.
- Calibration: reliability diagrams, ECE, empirical conformal coverage vs. nominal.
- Ablations: modality (IMU only / +gaze / +tool / +sEMG), policy components (remove mastery term; remove engagement latent; remove fairness constraint).
- Distribution-shift check: train early-session, test late-session, and report the degradation. This is the empirical statement of (P1).

13.3 Educational validation (does it teach?)

- **Concurrent validity:** system mastery estimate vs. blinded expert rubric scoring (checklist + global rating, OSATS-style structure). Two independent raters; report ICC and inter-rater reliability *before* reporting agreement with the system.
- **Predictive validity:** does mastery at session n predict independent performance at session $n+1$?

- **Learning-outcome comparison:** quasi-experimental, system-supported vs. instructor-only cohort, on steps-independent, time-to-criterion, error persistence, and retention at 4 weeks.
- **Usability/burden:** NASA-TLX for learners; instructor workload and trust measures.

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14. Results Analysis

Question	Method
Routing policy comparison	Repeated-measures ANOVA or linear mixed model, learner as random effect; Holm correction across the four baselines
Acquisition trajectory	Mixed-effects growth curve; learner-level random slopes; test for the predicted inverted-U in offload rate via quadratic term
Offload vs. mastery coupling	Within-learner correlation of offload rate against mastery, with autocorrelation correction
Agreement with rubric	ICC(2,1) for absolute agreement; Bland–Altman for systematic bias
Efficiency–fairness	Pareto frontier over offload budget; Jain's fairness index reported alongside aggregate gain
Learning gain	ANCOVA with pre-test as covariate; effect sizes with CIs, not p-values alone
Contention behaviour	Queueing analysis + simulation sweep over N = 10…40 learners

Reporting standard: effect sizes with confidence intervals throughout; pre-register the primary outcome (E2) and the analysis plan before the field trial begins; report all pre-registered outcomes including null results.

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15. Limitations

Stated proactively — a reviewer who finds these before you do will assume you were hiding them.

1. **Three tasks, one institution, one domain.** Generalisation beyond structured-cabling psychomotor skills is a claim this study cannot support.
2. **Motion does not imply correctness.** Mitigated by the dual-channel design, but outcome verification depends on tool instrumentation coverage; steps without an

instrumented outcome check are assessed on process quality alone and are weaker.

3. **Modest sample.** n≈30 learners limits detectable effect size for the learning-outcome comparison; the systems results are better-powered than the educational ones. Be explicit about which claims the sample supports.
4. **Quasi-experimental, not randomised.** Randomising instructional support within a graded practical raises equity concerns; expect a cohort-comparison design and control for prior attainment.
5. **Novelty effect.** Wearables in a lab are motivating in themselves. The 4-week retention measure partially addresses this; a longer follow-up would be better and is not budgeted.
6. **Eye-tracker ecological validity.** Tethered or bulky trackers alter natural head movement and may perturb the very behaviour being measured.
7. **Sensor placement variance.** Per-learner IMU calibration is required; drift across a long session is an open failure mode.
8. **Cloud FM dependency.** The offline customisation loop depends on an external API whose behaviour and availability are outside project control; version-pin and log all FM interactions.
9. **Instructor-authored task graphs** encode one institution's practice; a different lab's chaining order changes the skill graph and possibly the results.

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16. Risks and Mitigation

#	Risk	L	I	Mitigation	Trigge fallbac
R1	Ethics approval delayed past M2	M	**H**	Submit in week 1; prepare a no-video, IMU+tool-only protocol variant that is far easier to approve	No dec by end → swi reduce protoc
R2	Eye trackers unaffordable or unavailable	M	M	Phase gaze in at P3; IMU + tool channel is sufficient for E1–E3	Procur fails b → drop gaze, r as limi
R3	Mastery estimate fails E4 (r < 0.5)	M	**H**	Fall back to a coarse three-state stage classifier (cognitive/associative/autonomous)	r < 0.5 P3

#	Risk	L	I	Mitigation	Trigge fallbac
				— cruder, but still a valid conditioning signal	
R4	Insufficient learner enrolment / attrition	M	H	Recruit across both MMU practicals and the matriculation-college KTP cohort; over-recruit by 30 %	< 25 re by mid
R5	Trimester window missed	M	**H**	Treat the practical timetable as immovable; front-load P0–P4	Slip > 1 weeks end P4 descop simula only pa for the systen venue, field tr
R6	`B3` reproduction fails	L	H	Contact original authors early; if irreproducible, document and use a documented approximation, disclosed as such	Not wi 10 % b P0
R7	Instrumented tools unreliable	M	H	Redundant outcome verification via cable tester logs + photographic commitment-moment capture	> 10 % missin outcor labels
R8	Sensor time-sync drift corrupts multimodal fusion	M	H	PTP where possible; per-session sync beacon; automated integrity check that rejects sessions below threshold	> 5 % sessio rejecte
R9	Effect exists but is too small to detect	M	M	Power analysis at P1 using pilot variance; adjust design or reframe as a systems paper with an educational pilot	Pilot p < 0.7 f
R10	Scope drift back toward health/children/classroom	**H**	**H**	§5.2 is binding; any scope change requires re-running §11 timeline feasibility	Any propos add a domai

R10 is listed seriously. This proposal reached its current form after four scope changes; the cost of a fifth is a lost trimester.

17. Ethics and Data Governance

- **Approval:** MMU research ethics committee, submitted P0 week 1. Adults only in the current scope.
- **Consent:** written informed consent, separable by modality — a learner may consent to IMU and tool data but decline gaze or scene capture, and must remain in the class either way.
- **Assessment separation:** system output **must not** contribute to summative grades during the study. If learners believe the wearables are grading them, the behaviour under measurement is no longer the behaviour of interest, and the consent is no longer meaningfully voluntary.
- **PDPA compliance:** all identifiable data on-premise; pseudonymised at ingest; defined retention and destruction schedule; no third-party cloud storage of raw streams.
- **Bystander privacy:** scene cameras capture classmates. Scene frames terminate at Tier 1 (C1); lab signage; opt-out zone in the lab for non-participants.
- **Withdrawal:** any learner may withdraw at any time with retrospective data deletion, without academic consequence.
- **Federated updates:** cross-lab model improvement via DP-SGD with a stated and reported privacy budget (ε), not raw data pooling.

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18. Publication Plan

Target	Framing	Timing
Primary — systems (ACM IMWUT / IEEE IoT-J / IEEE TMC)	Mastery-conditioned model selection; contention scheduling; efficiency results E1–E3	M12
Secondary — education (IEEE ToLT / Computers & Education)	Wearable multimodal assessment of psychomotor acquisition; validity and learning outcomes E4–E6	M12+3
Dataset/demo	Annotated network-engineering psychomotor dataset; likely the most-cited output	With primary

Two papers, one system, non-overlapping claims. Do not attempt to place both contributions in one manuscript — the systems venue will not review the educational validation properly, and vice versa.

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19. Reference Status

Verified in preparation of this document:

Ref	Status
EdgeFM — *Leveraging Foundation Model for Open-set Learning on the Edge*, ACM SenSys 2023, arXiv:2311.10986	✅ Verified
DeepEdgeFM — *An Efficient Edge-Cloud Collaboration System With Foundational Models for Open-Set IoT Applications* (journal extension of EdgeFM)	✅ Verified — **obtain full text; exact venue/year to confirm**
SkillSight — *Efficient First-Person Skill Assessment with Gaze*, arXiv:2511.19629	✅ Verified — closest prior art, must differentiate
Generalized and Efficient Skill Assessment from IMU Data, ACM (10.1145/3422168)	✅ Verified — source of the repetition-consistency mastery proxy
Cengher et al. (2015), *Behavior Analysis in Practice* 9(2):115–125 — prompt-fading comparison	✅ Verified

Cited by the PI, NOT verified — do not use until checked:
`Nasar 2026` · `He 2025` · `Liu 2025` · `Li 2025` · `Fan 2023` — supplied as bare author-year with no titles. This pattern is characteristic of fabricated or auto-generated citation stubs. **Do not carry these into the manuscript without retrieving the full record.**

To be sourced during P0 literature review:
Fitts & Posner (1967) motor learning stages · BKT (Corbett & Anderson 1995) · DKT (Piech et al. 2015) · OSATS (Martin et al. 1997) · conformal prediction (Vovk / Angelopoulos & Bates tutorial) · calibration (Guo et al. 2017) · quiet-eye expertise literature (Vickers).

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20. Immediate Next Actions

1. **Submit ethics application.** Everything downstream is gated on it; nothing else is.
2. **Retrieve and read SkillSight and the DeepEdgeFM full text.** Write the differentiation paragraph before writing anything else.
3. **Verify or discard the five unconfirmed citations.**
4. **Book the practical-lab timetable for P5** against the trimester calendar and work the timeline backwards from it.

5. **Run the task analysis session** with instructors for T1–T3 and freeze the task graphs.
6. **Pre-register E2** as the primary outcome with its analysis plan.